Draft version October 25, 2026
Typeset using LaTeX **default** style in AASTeX7.0.1

A 1.6 $R_\oplus$ Transiting Planet Candidate on a 5.31-day Orbit around the Nearby M4.5 Dwarf G 249-11 (TIC 417732194)

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Keywords: Exoplanets (498) — Transit photometry (1709) — M dwarf stars (982)

1. INTRODUCTION

Small planets transiting nearby mid-M dwarfs are the most accessible rocky worlds for mass and atmospheric measurements. We report a shallow, short transit-like signal on G 249-11 (TIC 417732194; Gaia DR3 4846714643313496 32), an M4.5 dwarf at 29.09 pc with $T_{\rm eff} = 3180 \pm 157$ K, $R_\star = 0.266 \pm 0.008\,R_\odot$, $M_\star = 0.238 \pm 0.020\,M_\odot$ and $T_{\rm mag} = 12.30$ (Stassun et al. 2019; Gaia Collaboration 2023). The signal is not a TESS Object of Interest (TOI), community TOI, or SPOC threshold-crossing event, and no planet is known around the star.

2. DATA AND DETECTION

The candidate emerged from a box least squares (BLS; Kovács et al. 2002) search of TESS-SPOC full-frame-image light curves (Caldwell et al. 2020) of 7,474 M dwarfs within 40 pc. We then analysed all *TESS* (Ricker et al. 2015) photometry of the star: SPOC 2-minute PDCSAP light curves (Jenkins et al. 2016) from Sectors 19, 59 and 60, and QLP 200-s light curves (Huang et al. 2020) from Sectors 73 and 86. We detrended each sector with a 0.5-day biweight filter (Hippke et al. 2019) and clipped upward 4σ outliers (flares).

Discovery versus post-selection data. Sectors 19, 59 and 60 contributed to the detection and period selection. Sectors 73 and 86 were not used for selection. We fitted an ephemeris to Sectors 59–60 alone and predicted transit times in Sectors 73 (68 orbits later) and 86 (134 orbits later), searching only within the propagated $\pm 2\sigma$ timing window. Dips of the expected depth appear at the predicted times in both, with timing offsets (-1.2 and -2.9 h) that grow linearly with epoch. The final period predicts offsets of -1.4 and -2.8 h relative to the Sector 59–60 ephemeris, consistent with a single, slightly shorter period rather than independent noise dips.

To calibrate the significance we ran 400 end-to-end null realisations. In each, we masked the real transits, circularly shifted each sector's time series, and repeated the identical search and prediction chain.

- The candidate's Sector 59–60 search peak (SNR 9.2) exceeds all 400 null maxima (99th percentile 7.0).
- The post-selection signals are matched by chance with $p = 0.09$ (Sector 73) and $p = 0.0025$ (Sector 86). The combined probability is $p \approx 0.002$, about 3σ.

3. TRANSIT MODEL AND VETTING

We fitted all five sectors with a `batman` model (Kreidberg 2015) using `emcee` (Foreman-Mackey et al. 2013). We fixed quadratic limb darkening at (0.25, 0.35), appropriate for a mid-M dwarf in the *TESS* band (Claret 2017), assumed a circular orbit, and placed a Gaussian prior on the TIC stellar density ($12.7 \pm 2.5\,\rho_\odot$). The prior costs only $\Delta \ln L = 0.6$ relative to an unconstrained fit. We obtain:

- $P = 5.307425 \pm 0.000008$ d and $T_0 = 2459912.3151 \pm 0.0008$ BJD$_{\rm TDB}$;
- $R_p/R_\star = 0.055 \pm 0.003$, depth 3.0 ± 0.4 ppt, $T_{14} = 0.88 \pm 0.04$ h, $a/R_\star = 30 \pm 2$ and $b = 0.83 \pm 0.03$;
- $R_p = 1.61 \pm 0.09\,R_\oplus$, $a = 0.037$ au, $T_{\rm eq} \approx 410$ K (zero albedo) and $S \approx 4.8\,S_\oplus$.

Figure 1 shows the folded light curves. The combined detection significance is SNR ≈ 12 for white noise, or about 8 after inflating the errors for the excess event-to-event depth scatter (reduced $\chi^2 = 2.2$).

The signal passes the standard photometric tests:

- Odd and even depths agree (2.75 ± 0.46 vs. 3.49 ± 0.42 ppt, 1.2σ).
- There is no secondary eclipse at phase 0.5 (-0.13 ± 0.30 ppt), and no repeating dip at any other phase.
- The depth is stable (2.57–2.74 ppt) across seven detrending choices.
- Binned residuals at the transit timescale are white ($\beta = 0.90$–1.04).
- Searching the flipped light curve recovers nothing at the candidate's period.
- The star shows no significant rotation signal (Lomb–Scargle power ≤ 0.06) and only nine flares in five sectors.
- A Sector 59 `TESScut` (Brasseur et al. 2019) difference image places the flux deficit on or adjacent to the target pixel.

4. CAVEATS AND FOLLOW-UP

The candidate is not confirmed. Each sector alone is weak (SNR 3–7), and the evidence after selection amounts to about 3σ. Two astrophysical false positives remain open:

- An unresolved bound eclipsing companion. The unconstrained fit mildly prefers a denser eclipsed body, and Gaia reports RUWE = 1.20 with astrometric excess noise at 8.5σ.
- A Gaia source with $G = 17.2$, $20''$ from the target, about one *TESS* pixel away. It would need a $\sim$20% eclipse, and our single-pixel difference image cannot exclude it. Fainter sources at $24''$ and $38''$ are less plausible.

Brighter Gaia neighbours ($G = 15.3$ at $42''$) would require hours-long, near-total eclipses, so they are disfavored.

Several steps would test the candidate:

- Ground-based photometry in several bands at the predicted times. Seeing-limited imaging resolves the $20''$ neighbour, and a colour-independent depth would disfavor an eclipsing binary.
- High-resolution imaging to search for a close companion.
- Precise radial velocities. The expected semi-amplitude is about 3–4 $\mathrm{m\,s^{-1}}$ for a rocky composition.

The star is circumpolar from mid-northern latitudes. *TESS* will observe it again in Sector 125 (2027–2028, `tess-point`; Burke et al. 2020), which should yield $\sim$5 transits.

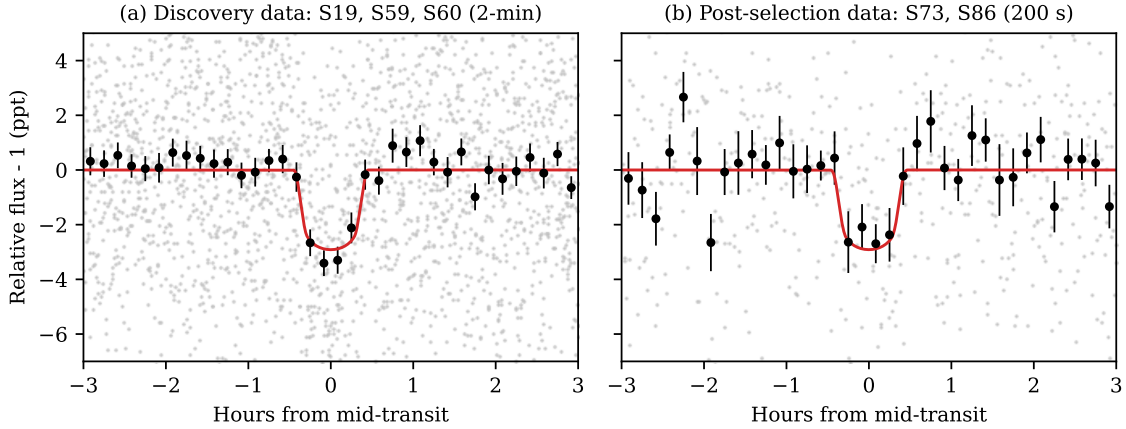


Figure 1. Detrended *TESS* photometry of G 249-11 folded on $P = 5.307425$ d. Gray points are unbinned and black points are 10-minute bins. The red curve is the density-prior transit model. (a) Discovery data: SPOC 2-minute, Sectors 19, 59 and 60. (b) Sectors 73 and 86 (QLP, 200 s), which were not used to select the candidate; they are folded on the final ephemeris, which includes them, and the blind prediction test is described in the text.

ACKNOWLEDGMENTS

The extracted photometry, fits, null tests, vetting report, figures, predicted transit times and analysis code are available on Zenodo (doi:10.5281/zenodo.23651677). This paper includes data collected by the *TESS* mission, obtained from MAST at STScI (doi:10.17909/t9-nmc8-f686, doi:10.17909/t9-r086-e880, doi:10.17909/t9-wpz1-8s54, doi:10.17909/0cp4-2j79). AI agents (a self-improving LLM research harness and Claude Code) carried out most of the search, the vetting and fitting code, and a first draft of this note. A separate agent session with read-only access reviewed the analysis adversarially and found errors that were corrected. The author directed the project, set the questions and acceptance criteria, checked the results, and takes responsibility for the content.

Facilities: TESS, Gaia

Software: astropy (Astropy Collaboration 2022), wotan (Hippke et al. 2019), batman (Kreidberg 2015), emcee (Foreman-Mackey et al. 2013), tess-point (Burke et al. 2020), TESScut (Brasseur et al. 2019)

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